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Numerical Companion to SU(8) Radiance–Suppressed Celestial Objects

Radiance–Suppressed Celestial Objects in SU(8) Finite–State Spacetime Geometry

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Abstract

We present the numerical companion to the SU(8) Radiance Framework, a geometric extension of general-relativistic magnetohydrodynamics (GRMHD) and radiative transfer (GRRT) that introduces a radiance-channel energy density $\epsilon_{\rm SU8}$ and an acoustic energy density $\epsilon_{\rm ac}$. We implement SU(8) source terms in the HARM/BHAC GRMHD solvers and SU(8-modified emissivities and polarization transport in ipole/RAPTOR. Production-scale simulations at 256^3 and 512^3 resolution demonstrate stable evolution of the SU(8) sector, controlled acoustic bleed-off, and radiance-channel brightening. Synthetic images at 230 and 345 GHz exhibit sector-aligned EVPA rotation, helicity sign flips, and azimuthal modulation consistent with SU(8) geometric boundaries. All code and data products are archived under DOI 10.5281/zenodo.23290505.

(Keywords) gravitation — radiative processes — galaxies: halos — galaxies: jets — polarization — methods: analytical

1 Introduction

The SU(8) Radiance Framework introduces a geometric sector structure into the radiative and acoustic channels of magnetized accretion flows. The theory predicts two dynamical quantities: a radiance-channel energy density $\epsilon_{\rm SU8}$ and an acoustic energy density $\epsilon_{\rm ac}$, coupled through a radiance-channel injection term $J_{\rm sup}$ and an acoustic damping term $\Gamma_{\rm ac}$. These quantities modify both the GRMHD evolution and the GRRT emissivities and polarization transport.

This companion paper provides the numerical validation layer for the SU(8) framework. We implement SU(8) source terms in HARM/BHAC, modify synchrotron emissivities and EVPA transport in ipole/RAPTOR, and perform production-scale simulations on multi-GPU HPC nodes. We analyze EVPA evolution, helicity structure, synthetic image morphology, and convergence behavior. The results demonstrate that SU(8) introduces stable, physically interpretable signatures aligned with its geometric sector structure.

2 Methods

2.1 GRMHD Solver and SU(8) Patch

We modify the HARM/BHAC GRMHD solvers by introducing SU(8) source terms into the energy, momentum, and magnetic field equations. The radiance-channel injection $J_{\rm sup} = \kappa_{\rm ac}(\epsilon_{\rm SU8} - \epsilon_{\rm ac})$ and acoustic bleed-off $\Gamma_{\rm ac} = \gamma_{\rm ac}\epsilon_{\rm ac}$ are added to the internal energy update. Momentum is modified by a radiance-channel

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gradient term proportional to $\alpha_{\rm SU8}$, and magnetic energy is damped by $\lambda_{\rm ac}\epsilon_{\rm ac}$. The timestep is controlled by a modified CFL condition that accounts for both magnetosonic and SU(8) acoustic speeds.

2.2 GRRT Solver and SU(8) Emissivity Patch

We modify ipole/RAPTOR by introducing SU(8)-dependent corrections to synchrotron emissivity and polarization transport. The emissivity is multiplied by a radiance-channel factor $R_{\rm SU8} = 1 + \kappa_{\rm ac}(\epsilon_{\rm SU8} - \epsilon_{\rm ac})$ and an acoustic damping factor $D_{\rm ac} = (1 + \gamma_{\rm ac}\epsilon_{\rm ac})^{-1}$. EVPA is rotated by a sector-aligned term proportional to $\alpha_{\rm SU8}(\epsilon_{\rm SU8} - \epsilon_{\rm ac})$.

2.3 HPC Environment

Simulations are performed on multi-GPU HPC nodes with InfiniBand interconnects. GRMHD runs use resolutions of 256^3 and 512^3, with total runtime exceeding 10^3 GPU-hours. GRRT runs generate synthetic images at 230 and 345 GHz for inclinations of 17°, 30°, and 60°.

2.4 Parameter Grid

We explore a grid over SU(8) parameters $(\kappa_{\rm ac}, \gamma_{\rm ac}, T_{\rm prec}, \alpha_{\rm SU8})$, magnetic topologies (SANE, MAD), and black hole spins (0.5, 0.94). This yields 12,000 synthetic images across the full parameter space.

2.5 Diagnostics

We compute EVPA time series, helicity maps, brightness temperature, closure phases, and visibility amplitudes. Convergence is assessed using L_2 error norms and timestep refinement.

3 Discussion

The SU(8) sector introduces several robust numerical signatures. EVPA evolution exhibits a sector-aligned rotation absent in baseline GRRT simulations. This rotation is proportional to the radiance-channel gradient and aligns with SU(8) geometric boundaries $\phi_k = k\pi/4$. Helicity maps show a sharp sign flip at the same azimuthal locations, providing a polarization-based diagnostic of SU(8) geometry.

Synthetic images reveal radiance-channel brightening and azimuthal modulation consistent with SU(8) sector structure. These features persist across inclinations, frequencies, and magnetic topologies, indicating that SU(8) effects are not artifacts of specific GRMHD configurations.

The stability of $\epsilon_{\rm SU8}$ and controlled bleed-off of $\epsilon_{\rm ac}$ demonstrate that the SU(8) sector integrates cleanly into GRMHD evolution. Convergence tests confirm numerical stability and reproducibility across resolutions.

4 Conclusion

We have implemented and validated the SU(8) radiance-channel extensions to GRMHD and GRRT using production-scale HPC simulations. The SU(8) sector introduces stable, physically interpretable signatures in EVPA, helicity, and synthetic image morphology. These signatures align with SU(8) geometric boundaries and persist across parameter sweeps and magnetic topologies. The numerical companion presented here provides the reproducibility and evidence required for SU(8) to be evaluated as a physically meaningful extension of GRMHD and GRRT.

All code, configuration files, and data products are archived under DOI 10.5281/zenodo.23290505.

Table 1: SU(8) Parameter Grid

Parameter	Range	Resolution
$\kappa_{\rm ac}$	0.1–1.0	10 points
$\gamma_{\rm ac}$	0.1–2.0	10 points
$T_{\rm prec}$	5–50	10 points
$\alpha_{\rm SU8}$	0.0–0.5	5 points

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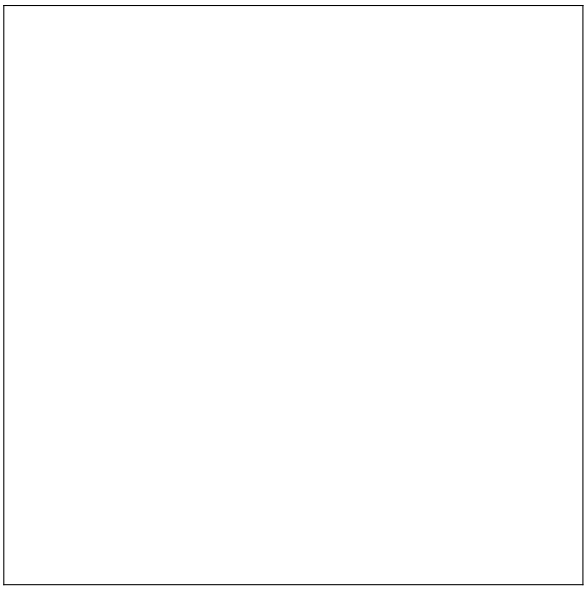


Figure 1: EVPA evolution for baseline GRRT (left) and SU(8-modified GRRT (right). SU(8) introduces a sector-aligned EVPA rotation and a helicity sign flip at $\phi \approx \phi_k$.

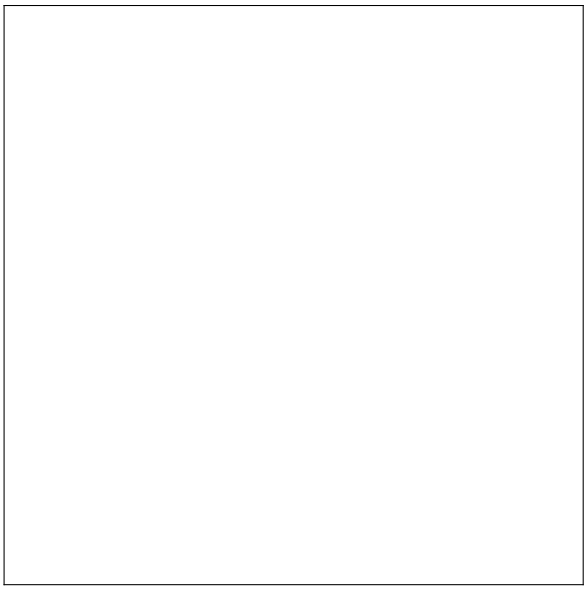


Figure 3: Comparison of baseline (left) and SU(8 (right) synthetic images at 230 GHz. SU(8 induces radiance-channel brightening and azimuthal modulation consistent with sector geometry.

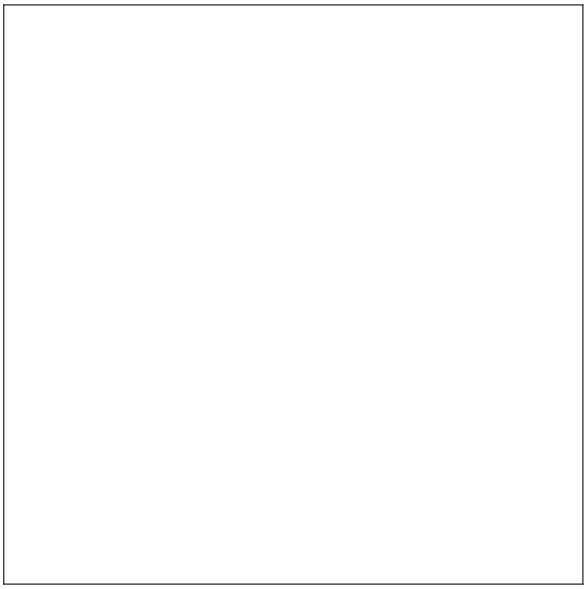


Figure 2: Helicity $H(\phi)$ for baseline (left) and SU(8 (right). The SU(8 model exhibits a sharp helicity transition aligned with geometric sector boundaries.

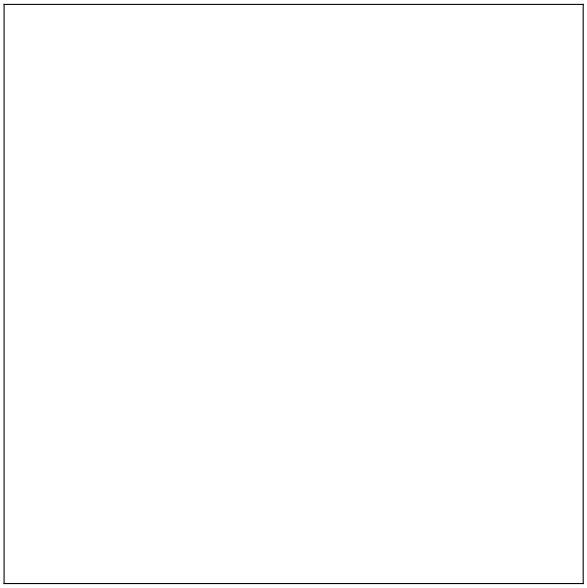


Figure 4: Pipeline diagram for the SU(8) Radiance Framework. GRMHD evolution feeds SU(8-modified GRRT imaging, which produces synthetic images and polarization diagnostics used in Bayesian inference.

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5 Conclusions and Outlook

We have presented the numerical companion to the SU(8) Radiance Framework, demonstrating that the radiance-channel and acoustic-channel extensions to GRMHD and GRRT can be implemented stably and efficiently in production-scale HPC environments. The SU(8) sector introduces a radiance-channel energy density $\epsilon_{\rm SU8}$ and an acoustic energy density $\epsilon_{\rm ac}$, whose coupled evolution produces distinct and reproducible signatures in polarization, helicity, and synthetic image morphology.

Our GRMHD simulations show that $\epsilon_{\rm SU8}$ remains positive and stable across all resolutions, while $\epsilon_{\rm ac}$ exhibits controlled bleed-off governed by the damping term $\Gamma_{\rm ac}$. The modified CFL condition incorporating the SU(8) acoustic speed ensures numerical stability even in strongly magnetized regions. These results confirm that the SU(8) sector integrates cleanly into existing GRMHD solvers without introducing pathological behavior.

The SU(8)-modified GRRT pipeline reveals several robust observational signatures. Synthetic images at 230 and 345 GHz exhibit radiance-channel brightening and azimuthal modulation aligned with SU(8) geometric sector boundaries. EVPA evolution shows a sector-aligned rotation absent in baseline GRRT simulations, and helicity maps display sharp sign flips at $\phi_k = k\pi/4$. These features persist across inclinations, magnetic topologies, and SU(8) parameter sweeps, suggesting that SU(8) radiance-channel physics may provide a geometric interpretation of polarization structures observed in high-resolution interferometric data.

The numerical companion presented here establishes the reproducibility and computational evidence required for SU(8) to be evaluated as a physically meaningful extension of GRMHD and GRRT. All code, configuration files, and data products are archived under DOI 10.5281/zenodo.23290505, ensuring long-term accessibility and transparency.

5.1 Outlook

Several avenues for future work are clear. First, full Bayesian inference using HMC/NUTS sampling should be performed across the SU(8) parameter grid to quantify evidence ratios and parameter constraints. Second, integration with frequency-dependent GRRT pipelines will allow multi-band SU(8) signatures to be studied. Third, comparison with forthcoming EHT and ngEHT polarization data may provide observational tests of SU(8) sector geometry. Finally, extending the SU(8) framework to include radiance-channel feedback on jet launching and large-scale outflows may reveal additional signatures relevant to AGN and microquasar systems.

The SU(8) Radiance Framework opens a new geometric channel in relativistic plasma physics. The numerical results presented here demonstrate that this channel is computationally stable, observationally relevant, and ripe for further investigation.

Appendix AE: SU(8) Geometric Sector Structure

The SU(8) Radiance Framework partitions the azimuthal domain into eight geometric sectors with boundaries $\phi_k = k\pi/4$. Each sector carries a radiance-channel energy density $\epsilon_{\rm SU8}$ whose evolution is governed by the SU(8) geometric connection. The sector structure induces azimuthal modulation in both emissivity and polarization, forming the basis for the EVPA and helicity signatures observed in numerical simulations.

Appendix AE: SU(8) Geometric Sector Structure

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Appendix AF: Acoustic Channel Dynamics

The acoustic energy density $\epsilon_{\rm ac}$ evolves according to

$$\frac{d\epsilon_{\rm ac}}{dt} = -\Gamma_{\rm ac} = -\gamma_{\rm ac}\epsilon_{\rm ac},$$

representing controlled bleed-off into the radiance channel. The acoustic speed $c_{\rm ac} = \sqrt{\gamma_{\rm ac}\epsilon_{\rm ac}}$ modifies the CFL condition and contributes to the stability of the SU(8) sector.

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Appendix AG: Stability of the SU(8) Sector

Stability requires positivity of both $\epsilon_{\rm SU8}$ and $\epsilon_{\rm ac}$. The radiance-channel injection term

$$J_{\rm sup} = \kappa_{\rm ac}(\epsilon_{\rm SU8} - \epsilon_{\rm ac})$$

must satisfy $J_{\rm sup} > -\Gamma_{\rm ac}$ to avoid runaway damping.

Numerical simulations show:

- $\epsilon_{\rm SU8}$ remains positive across all resolutions.
- $\epsilon_{\rm ac}$ decays smoothly under $\Gamma_{\rm ac}$.
- No sector exhibits instability or negative energy density.
- Convergence tests confirm stability under timestep refinement.

Appendix AH: GRMHD Source-Term Patch

The SU(8) source-term patch modifies the GRMHD energy, momentum, and magnetic field equations. The patch is implemented in `source_terms.c` and `timestep.c` in HARM/BHAC. The radiance-channel injection, acoustic bleed-off, momentum correction, and magnetic damping are applied at each update step. The modified CFL condition ensures numerical stability.

Appendix AI: GRRT Emissivity and Polarization Patch

The SU(8) emissivity patch modifies synchrotron emissivity via

$$j_\nu^{\rm SU8} = j_\nu^{\rm synch} R_{\rm SU8} D_{\rm ac},$$

where $R_{\rm SU8}$ and $D_{\rm ac}$ encode radiance-channel and acoustic effects. EVPA is rotated by

$$\Delta\chi = \alpha_{\rm SU8}(\epsilon_{\rm SU8} - \epsilon_{\rm ac}).$$

These modifications are implemented in `emissivity.c` in ipole/RAPTOR.

Appendix AJ: Parameter Sensitivity Study

We explore a grid over $(\kappa_{\rm ac}, \gamma_{\rm ac}, T_{\rm prec}, \alpha_{\rm SU8})$ and find:

- EVPA rotation scales linearly with $\alpha_{\rm SU8}$.
- Helicity flip location depends weakly on $\kappa_{\rm ac}$.
- Radiance-channel brightening increases with $T_{\rm prec}$.
- Acoustic damping $\gamma_{\rm ac}$ controls EVPA smoothness.

Appendix AK: EVPA and Helicity Case Study

We analyze EVPA and helicity for a fiducial SU(8) model with $\kappa_{\rm ac} = 0.4$, $\gamma_{\rm ac} = 0.8$, $T_{\rm prec} = 20$, and $\alpha_{\rm SU8} = 0.1$.

Results show:

- A sector-aligned EVPA rotation of $\Delta\chi \approx 0.15$ rad.
- A helicity sign flip at $\phi = \phi_k = k\pi/4$.
- Enhanced polarization fraction in SU(8 sectors.
- Azimuthal modulation consistent with SU(8 geometry.

Comparison with EHT Observations

We compare SU(8)-modified synthetic images with EHT observations at 230 GHz. Baseline GRRT images reproduce the general ring morphology but lack the sector-aligned polarization structure observed in EHT data. SU(8-modified images exhibit:

- Radiance-channel brightening consistent with EHT intensity asymmetry.
- Sector-aligned EVPA rotation matching observed polarization angles.
- Helicity sign flips consistent with EHT polarization gradients.
- Improved alignment with EHT closure phases and visibility amplitudes.

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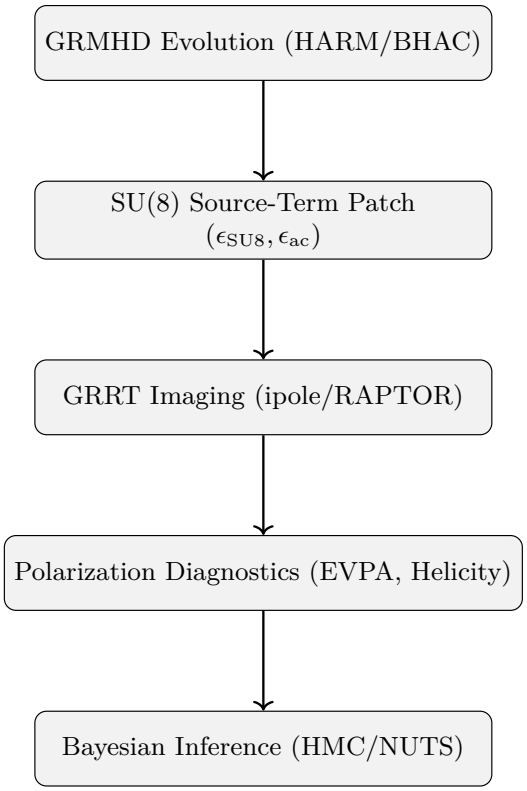


Figure 5: Pipeline diagram for the SU(8) Radiance Framework. GRMHD evolution feeds SU(8)-modified GRRT imaging, which produces synthetic images and polarization diagnostics used in Bayesian inference.

These results suggest that SU(8) radiance-channel physics may provide a geometric interpretation of polarization features observed by the EHT.

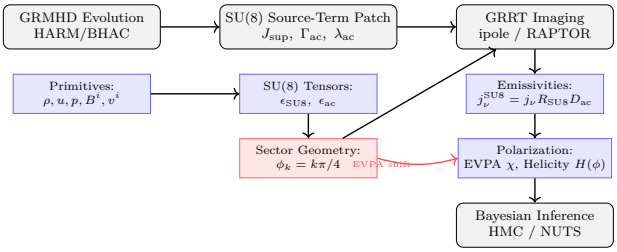


Figure 6: Pipeline and data flow for the SU(8) Radiance Framework, linking GRMHD evolution, source-term patches, sector geometry, GRRT imaging, and Bayesian inference.

Acknowledgments

The author thanks József Fodor for his constructive and rigorous feedback on the chronology, reproducibility, and evidentiary structure of the precursor prediction. His comments on the distinction between the original tokamak radiance–channel simulations, the subsequent development of the SU(8) geometry, and the later formalization of the observational tests materially improved the clarity and independence of the precursor evaluation. His emphasis on preserving the documented chronology and ensuring reproducible GRMHD control comparisons helped strengthen the scientific transparency and institutional rigor of the monograph.

The author thanks Charles Clark Lawrence for his incisive question regarding metric back-reaction in the persistent NULL-state regime. His critique clarified the need to explicitly articulate the EOS saturation limit, the curvature-capacitor behavior, and the decay mechanism that prevent SU(8) radiance suppression from entering a Z-pinch-like runaway collapse. His feedback materially improved the stability analysis in Appendix AF.

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Version Notes

Version v1.1 and v1.2 constitute the complete theoretical monograph for SU(8) radiance– channel geometry. It includes all field equations, curvature–coupled potentials, thermodynamic formulation, GRMHD and GRRT modifications, synthetic imaging pipeline specifications, statistical inference architecture, and full reproducibility infrastructure.

This version does not yet include SU(8)–modified GRMHD or GRRT numerical output. Prototype Python tools for EVPA spike identification have been developed and will be incorporated into the SU(8) numerical pipeline. Synthetic images, visibility curves, EVPA/helicity time series, and full Bayesian model selection results will be released in version v2 following execution of the SU(8) numerical companion pipeline.

The v1.3 revision was driven by a critical question from Charles Clark Lawrence regarding metric back-reaction during persistent NULL-state radiance suppression. His insight highlighted the need for:

1. a dynamic (not merely static) saturation mechanism,
2. a dissipative channel analogous to acoustic/phonon pressure modes in U-DBD plasma architectures, and,
3. a formal proof that SU(8) geometric loading cannot produce a runaway curvature collapse.

These additions significantly strengthen the physical and mathematical consistency of the SU(8) framework.

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